

Resume

Skills

DOCKER

PYTHON

TERRAFORM

JAVA

C

CMAKE

JAVASCRIPT

AMAZON WEB SERVICES (AWS)

SOFTWARE ARCHITECTURE

REST

GRAPHQL

MICROSERVICES

NETWORKING

CI/CD

AGILE

DATABASE DESIGN

DJANGO

CELERY

RABBITMQ

REDIS

FASTAPI

MYSQL

POSTGRES

CYBERSECURITY

AUTOMATION

LINUX

GIT

SHELL SCRIPTING

SYSTEMS ADMINISTRATION

REACT

HTML

CSS

Work Experience

Elucid Bioimaging

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Summary

(<https://elucid.com/>).

Title	Principal Platform Engineer	As Elucid's Principal Platform Engineer, I lead the architecture, design, and implementation of the cloud platform that powers the company's arterial analysis software. My responsibilities span platform infrastructure, release engineering, and site reliability — guiding the evolution of a regulated medical-imaging product from desktop application to cloud-native SaaS.
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Platform Architecture

Responsibilities:

- lead the design and implementation of an AWS ECS (<https://aws.amazon.com/ecs/>)-based containerization platform for the company's arterial analysis software, reducing production deployment time from hours to under five minutes
- architect a cross-region warm-standby disaster recovery footprint, defining the Terraform IaC (<https://developer.hashicorp.com/terraform/tutorials/aws-get-started/infrastructure-as-code>), patterns for network, compute, storage, and identity layers
- design asynchronous task-processing infrastructure for backend services, introducing managed message brokering and dedicated worker tiers to support scalable, long-running workloads
- guide the evolution of the company's service-based architecture, partnering with engineering leaders to navigate organizational and technical trade-offs
- establish cross-account, multi-region distribution patterns for AMIs (<https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/AMIs.html>), and container images, decoupling build pipelines from deployment topology
- own multi-account cost management for over eighty environments serving six product development teams, balancing engineering velocity with budget accountability

Site	elucid.com (https://elucid.com/)
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DevOps Leadership

Responsibilities:

- own monthly release cadence across multiple coupled product repositories, coordinating cross-team delivery and the supporting regulatory documentation
- design and build internal CLI tooling for orchestrating multi-repository releases, consolidating configuration, and enabling developer self-service workflows
- democratize backend service development by delivering per-team containerized development environments, accelerating feature parallelism across the engineering organization
- lead adoption of agile delivery practices including backlog refinement, sprint retrospectives, and architectural decision records
- mentor engineers across teams through design reviews, pull request feedback, and pairing on platform-engineering tasks

Site Reliability & On-Call

Responsibilities:

- establish the company's on-call program, defining service-level indicators and objectives, runbook standards, and a blameless postmortem culture
- implement Slack-integrated alerting tooling to streamline incident triage and reduce time-to-acknowledge
- design observability dashboards spanning deployment pipeline health, environment version tracking, and infrastructure performance
- lead adoption of application performance monitoring  [\(https://docs.datadoghq.com/tracing/\)](https://docs.datadoghq.com/tracing/) and software cataloging to clarify service ownership across the engineering organization
- partner with engineering teams to embed reliability practices into the regular development lifecycle

Hometap



hometap

Summary

[\(https://www.hometap.com/\)](https://www.hometap.com/)

TitleDevOps Engineer

My position as Hometap's first devops engineer provided a unique opportunity to help guide the team through the challenges of its devops evolution. I was responsible for the design, implementation, and support of the company's cloud infrastructure, CI/CD pipelines, automation tooling, and application monitoring systems.

Start

2021 November

Platform Engineering

Responsibilities:

- design the cloud architecture and implementation plan of an AWS ECS (<https://aws.amazon.com/ecs/>)-based platform for running containerized backend services
- lead the migration from disparate code repositories with inconsistent support for CI/CD to a unified, standardized development platform providing a predictable and consistent developer experience
- collaborate cross-functionally with internal stakeholders to identify pain points and bottlenecks between scrum teams, and develop solutions to address them
- implement, maintain, and champion cross-platform Docker image builds  (<https://docs.docker.com/build/building/multi-platform/>) as engineering team transitioned from x86_64 to arm64 development machines

Site	hometap.com (https://www.hometap.com/)
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DevOps Engineering

Responsibilities:

- design and implement CI/CD pipelines for multiple applications, languages, and life cycles using the following platforms: GitLab CI/CD  (<https://docs.gitlab.com/ee/ci/pipelines/>), GitHub Actions  (<https://docs.github.com/en/actions>), CircleCI (<https://circleci.com/>).
- optimize the execution speed of a large Python / Django (<https://www.djangoproject.com/>) test suite, reducing lead time for changes (<https://www.atlassian.com/devops/frameworks/devops-metrics>) and change failure rate (<https://www.atlassian.com/devops/frameworks/devops-metrics>).
- guide the team's transition to using Terraform IaC (<https://developer.hashicorp.com/terraform/tutorials/aws-get-started/infrastructure-as-code>) for consistent creation and management of cloud infrastructure across multiple AWS accounts
- implement and maintain Terraform IaC (<https://developer.hashicorp.com/terraform/tutorials/aws-get-started/infrastructure-as-code>) stacks for managing cloud resources in the following AWS services: Amplify (<https://aws.amazon.com/amplify/>), CloudFront (<https://aws.amazon.com/cloudfront/>), CloudWatch (<https://aws.amazon.com/cloudwatch/>), CloudTrail (<https://aws.amazon.com/cloudtrail/>), EC2 (<https://aws.amazon.com/ec2/>), ECR (<https://aws.amazon.com/ecr/>), ECS (<https://aws.amazon.com/ecs/>), IAM (<https://aws.amazon.com/iam/>), RDS (<https://aws.amazon.com/rds/>).

Title	Full-Stack Software Developer	As a senior developer on a rapidly growing team, I was able to provide guidance and technical insight to younger developers. Through effective collaboration, troubleshooting, process development, and technical accomplishment, I established a reputation as a subject matter expert for Docker, Python, Terraform, AWS, <code>git</code> , GitLab, CI/CD, DevOps, and cloud-native web applications.
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Start

2020 August

Full-Stack Software Developer

In this position, I met the following responsibilities:

- Provide in-depth expertise for DevOps best practices and administration, supporting the cloud deployment, operations, and scaling of an internally-managed GitLab  [\(https://about.gitlab.com/\)](https://about.gitlab.com/) instance
- Implement advanced CI/CD pipelines for automating deployments of a microservices-based web application across multiple staging and production environments
- Implement automation tooling for quantifying QA metrics, improving product stability, development speed, and developer satisfaction
- Implement Terraform IaC for scalable, reusable cloud deployment configurations of AWS services to support internal product development and operations

Site	morsecorp.com (https://www.morsecorp.com/)
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Outdoorlink

Summary

Start2020 August

I was the primary contributor to the development and maintenance of a distributed web application providing Django-based REST APIs and a React-based JavaScript frontend hosted in an AWS VPC (<https://aws.amazon.com/vpc/>), with redundancy managed through Celery (<https://docs.celeryproject.org/en/stable/index.html>), and AWS RDS (<https://aws.amazon.com/rds/>), for increased availability. The application was designed to manage scheduling, status, communications, and operations of an IoT fleet of utility power management devices utilizing cellular communication technologies for the control of more than 70,000 billboard structures in the US, Australia, and South America.

End	2022 January
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After relocating to Boston, MA, I retained a part-time consulting position at Outdoorlink.

Site

smartlinkcontrol.com
(<https://smartlinkcontrol.com>).

Consulting

- Contributed to a new product launch by kick-starting a RESTful API for modeling and managing product data
- Developed a Django application for user identity and access management

AWS Infrastructure

I supported maintenance and operations of the infrastructure that Outdoorlink uses to host its web applications and data. Additionally, I implemented Terraform IaC (<https://www.terraform.io/intro/index.html>) for deploying new infrastructure components and services.

- Maintained and enhanced GitLab CI/CD pipelines for application deployment and updates
- Implemented GitLab CI/CD pipelines for consistent deployments of Terraform-managed resources
- Used Terraform to enhance cloud security through high-precision security group and networking configurations
- Developed new deployment strategies to minimize infrastructure costs and downtime while improving scalability

DevOps Engineering

- Pushed testing left by developing multistage CI/CD pipelines for automating the following tasks:
 - code static analysis, linting, formatting, style checks, test suite execution, and coverage reporting
 - dependency updates, package installation, runtime image builds, documentation builds, and deployments to test and production environments
- Maintained internal GitLab 🏠 (<https://about.gitlab.com/>) server for hosting project repositories, collaborating on feature requests and bug fixes, and reviewing code changes and deployments
- Containerized applications with Docker 🐳 (<https://www.docker.com/why-docker>) to support concurrent test suite executions across on-prem servers, reducing the feedback cycle for testing requested changes to the codebase

Full-Stack Web Development

- Supported development and migration from a legacy SOAP API to a modern Django-based REST API for manipulating IoT schedule settings
 - Communicated with customers for identifying API requirements and prioritizing feature requests/bug fixes
 - Created URL schema documentation with Swagger (<https://portal.outdoorlinkinc.com/thirdparty/swagger/>), Redoc (<https://portal.outdoorlinkinc.com/thirdparty/redoc/>), and Postman (<https://documenter.getpostman.com/view/6606680/SVYjU3Kd>)
 - Developed reusable Django application for logging REST API HTTP request details to multiple servers over TCP, supporting better OutdoorLink support to customers utilizing the REST API
- Supported app migration from on-prem servers to an AWS cloud deployment
 - Managed AWS infrastructure for networking, compute resources, user authorization, scaling, and deployments
 - Configured system and application logging across AWS EC2 (<https://aws.amazon.com/ec2/>) instances for `NGINX`, `python`, and `syslog` for improved visibility and troubleshooting
- Integrated sentry.io (<https://sentry.io/welcome/>) for error monitoring, tracking, and triage
- Optimized PostgreSQL DBMS configuration to mitigate OOM conditions, stale connections, and disk space usage

End	2020
	August

IERUS Technologies

Title

Engineer

Summary

Start	2016 October
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As an engineer at IERUS Technologies, I contributed to a variety of SBIR contracts for the Department of Defense and similar research initiatives. I also worked on-site at Raytheon for several subcontracts. My contributions included software development, radar processing/analysis, and modeling/simulations.

Conformal Conductivity Probe

- Developed Python application for interactively controlling a vector network analyzer used to measure the condition of aircraft skin joints
 - Utilized Python's QT package for displaying data in realtime as RF measurements were taken
 - Created a Python connector to interface with the VNA's COM server
 - Created an emulator for the VNA's COM server to allow cross-platform development and testing until the COM-based drivers were replaced

DASH

- Developed shared C++ libraries for parsing XML configuration files and integrated the parser with the rest of the trajectory propagation application
 - Designed XML schema for storing trajectory 6 DoF satellite trajectory data
 - Used OpenCL binaries to reference GPU kernel modules to increase speed of runge-kutta derivative calculations

Third-party Tracker

- Developed test framework for verifying C++ implementations of different Kalman filter models
 - Used MatLab to build tracker program from C++ with autotools
 - Test cases in the framework ran the filter implementations on specified test data sets and then generated summary figures for quickly reviewing builds
 - Platform independence allowed the framework to be used as a manual build server

Methodologies of Accurate Assessment of Target Characteristics

- Analyzed the effectiveness of different machine-learning techniques and Artificial Neural Network (ANN) designs when used to process radar measurement data
 - Used the scikit-learn (<https://scikit-learn.org/stable/>). Python package for training the ANNs from CentOS machines.
 - Created visualizations of ANN performance when classifying and labeling data according to the target feature space
 - Prototyped parallelization of ANN training source using bash scripts to mimic parallelized computing in Python

RMCI, Inc.

Summary

Start 2012 August	After interning with RMCI during my senior year of college, I grew into a full-time role as embedded systems engineer. I contributed to the development of <u>XRDS™</u> (https://www.rmci-inc.com/our-system/), a single-board computer system that measured vibration signals from helicopter drive train components to detect hardware faults.
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End

2016
October

XRDS™ Software Development

- Developed and tested vibration data analysis algorithms for condition-based maintenance from an embedded OS
 - Implemented the algorithms in C for utilization by an ARM cortex 335x series (https://www.ti.com/lit/ds/symlink/am3359.pdf?ts=1634308387318&ref_url=https%253A%252F%252Fwww.google.com%252F) microcontroller. The focus of the algorithms was on time domain, frequency domain, and synchronous domain analyses
 - Developed a high-level framework in C incorporated with the Yocto project (<https://www.yoctoproject.org/>) for handling onboard processing
 - Designed and integrated communication functions with an embedded SQLite (<https://www.sqlite.org/index.html>) database for retrieving algorithm parameters, as well as loading and storing both raw and processed data
 - Interfaced with low-level routines for collecting raw data and monitoring system state via a watchdog timer
 - Established communications protocol for system-wide integration in order to perform all onboard data acquisitions and processing upon triggers either by the user or by flight regime
 - Performed software verification testing by comparing results from the data analysis framework with a Python transliteration of the original MatLab scripts. The results were displayed and analyzed

- Designed and tested firmware for the PIC24F08KL301 MCU using MPLAB X, Eclipse, and PICKIT3
 - The microcontroller was used to emulate tachometer signals with two 8-bit timers and to translate between RS-485 and RS-232 message protocols for interfacing with a rotor blade tracking system
 - The two UART peripherals were utilized for the serial communications: one dedicated to RS-485 communication with the controlling OS, and the other was dedicated to RS-232 communication with the tracking system
 - Implemented a communications protocol for responding to various control bytes transmitted over RS-485
 - Developed a boot loader for executing firmware updates over serial traffic: created host side in Python using minimal packages for ease of deployment on the embedded Linux system, and tested the MCU side written in C
 - Phase 1 development was done on the SAMD21G18A MCU with Eclipse and the Arduino IDE for programming

Rotor Blade Tracking System Design

- Led a multidisciplinary team to design a new product for tracking helicopter rotor blades
 - Performed individual research and bench-marking, theoretical modeling using linear algebra and geometry, and data visualization for demonstrating theoretical system performance characteristics. Provided the recommendation for the selected design and continued to lead the development of this design
 - Conducted electro-optical radiometric modeling of solid state sensors for blade detection
 - Completed optical design of system hardware using matrix ray tracing in MathCAD, Python, and Matlab
 - Developed opto-mechanical design of illuminator and sensor to minimize space requirements while considering reliability, affordability, and maintainability
 - Reviewed materials and manufacturing processes for component design and selection
 - Design criteria included the following: overall size, weight, and cost; IR safety compliance according to IEC 60825-1 and IEC 62471 ; insensitivity to environmental lighting conditions; environmental qualification according to D0-160 and MIL-STD-810 ; hardware reliability, aesthetic appeal, and convenience of operation to the end user
 - The system is currently entering the prototype stage and is estimated to cost less than existing systems while improving performance and reducing maintenance costs

- Performed all engineering analysis, design, and fabrication of a stand for testing developmental systems
 - The stand replicated a scaled rotor of approximately 8 ft. diameter, capable of rotational speed over 300 rpm
 - Used finite airfoil drag analysis to model the transient and steady-state speed based on input power. Total cost was under \$1,000 , and the test stand was used for rotor track and balance tests

Rotor Blade Tracking System Integration

- Developed and integrated shared libraries in C for communication and control of third-party tracker system
 - Libraries utilized two communications channels over RS-485 and RS-232 for configuration transmission and data retrieval
 - Protocols included ASCII and non-ASCII binary packets with checksums and ASCII hex representations
 - Configuration parameters were retrieved from SQLite Database and ASCII configuration file
 - Developed and integrated libraries for tokenizing and parsing ASCII file contents with comments

- Integrated tracker communications, acquisition, and processing libraries with framework for collecting RTB data
 - Developed lightweight logging library using variadic functions for creating and writing to log files, as well as writing messages to `stdout` for real-time feedback during developmental testing of framework libraries
 - Assisted with aircraft installation and testing on the airframes `OH-58C` and `OH-6A` of the entire embedded system in aspects of vibration data collections and rotor track height data collections

University of Alabama in Huntsville

Summary

Title

Research Assistant

I worked part-time as a research assistant at the Systems Management and Production (SMAP) Center (<https://www.uah.edu/smap>), where I contributed to the development and operations of small unmanned aerial vehicles.

Research Assistant

- Designed a mechanism for releasing non-lethal grenades from multirotor UAVs to assist SWAT teams
- Built, repaired, and maintained quadcopter and hexacopter UAVs as part of a team. Used SolidEdge to design replacement parts, and integrated imaging payloads with existing systems
- Wrote code for the Evr2est project, which is a system that processes and disseminates imagery from satellites or aircraft for government use in first response and wartime scenarios
- Worked on the design of a VTOL UAV for image acquisition; contributed to CAD modeling, stress analysis, and aerodynamics calculation

End

2012 December

Site	uah.edu/smap (https://www.uah.edu/smap).
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Title

PASS Leader

Summary

Start2010 August

I lead group study sessions for students of all ability levels in the following courses: Differential and Integral Calculus; Engineering Mechanics: Statics; and Engineering Mechanics: Dynamics.

PASS Leader

- Formed group study sessions for students of all levels of ability in the following classes: Differential and Integral Calculus; Engineering Mechanics: Statics; and Engineering Mechanics: Dynamics
- Organized and administered group study sessions, engaging students reinforcing understood class material and teaching class material that was not understood
- Derived representative practice problems for test preparation
- Identified and communicated fundamental course concepts to foster student learning
- Individually tutored students in engineering, math, and physics classes

Site	uah.edu/ssc/pass (https://www.uah.edu/ssc/pass)
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Education

**The University of Alabama in
Huntsville**

**2009 -
2013**

GPA

4.00

Security+ Certification

***April
2021***

Issued by CompTIA.org (<https://comptia.org>).

 **Updated:** May 28, 2026